1 **(a)** Arc length $BD = \frac{1}{4} \times 2 \times \pi \times 10 = 5\pi$ **[M1]**

Perimeter $= 10 + 10 + 5\pi = 20 + 5\pi$ **[A1]**

(b) Area of quarter circle $= \frac{1}{4} \times \pi \times 10^2 = 25\pi$ **[M1]**

Area of shaded region $= 100 - 25\pi$ **[A1]**

[Total Marks: 4]

2 **(a)** Arc length $= \frac{1}{2} \times \pi \times 10$ $(= 5\pi)$ **[M1]**

Perimeter $= 12 + 12 + 10 + 5\pi$ **[M1]**

$34 + 5\pi$ **[A1]**

(b) Area of rectangle $= 12 \times 10$ $(= 120)$ **[M1]**

Area of semicircle $= \frac{1}{2} \times \pi \times 5^2$ $(= 12.5\pi)$ **[M1]**

$120 + 12.5\pi$ **[A1]**

[Total Marks: 6]

3 **(a)** Volume of large cone $= \frac{1}{3} \times \pi \times 8^2 \times 24$ $(=512\pi)$ **[M1]**

Volume of small cone $= \frac{1}{3} \times \pi \times 5^2 \times 15$ $(=125\pi)$ **[M1]**

1215.8 (or 1220 or 387π) **[A1]**

(b) Slant height of large cone $= \sqrt{8^2 + 24^2}$ $(=8\sqrt{10}$ or $25.298\ldots)$ **[M1]**

Slant height of small cone $= \sqrt{5^2 + 15^2}$ $(=5\sqrt{10}$ or $15.811\ldots)$ **[M1]**

Curved SA of frustum $= \pi \times 8 \times 8\sqrt{10} - \pi \times 5 \times 5\sqrt{10}$ $(=387.4\ldots)$ **[M1]**

Total SA = Curved SA $+ \pi(8^2) + \pi(5^2)$ **[M1]**

667 (or $667.0\ldots$) **[A1]**

[Total Marks: 8]

4 **(a)** Identifies that the centers form an equilateral triangle of side $2r$ **[M1]**

Area of triangle $= \frac{1}{2} \times 2r \times 2r \times \sin(60°) = r^2\sqrt{3}$ **[M1]**

Identifies 3 sectors inside the triangle, each with angle 60°: Total sector area $= 3 \times \frac{60}{360} \times \pi r^2 = \frac{\pi}{2} r^2$ **[M1]**

Area $= r^2\sqrt{3} - \frac{\pi}{2} r^2 = r^2 \left(\sqrt{3} - \frac{\pi}{2}\right)$ with no errors seen **[A1]**

(b) $r^2 \left(\sqrt{3} - \frac{\pi}{2}\right) = 15$ **[M1]**

$r^2 = \frac{15}{\sqrt{3} - \pi/2}$ (≈ 93.02) **[M1]**

9.64 (or 9.644...) **[A1]**

(c) Perimeter consists of 3 arcs, each with angle 60° **[M1]**

Total perimeter $= 3 \times \frac{60}{360} \times 2\pi(9.644\ldots) = \pi(9.644\ldots)$ **[M1]**

30.3 (or 30.29 to 30.30) **[A1]**

[Total Marks: 10]

5 (a) Volume of cylinder $= \pi \times 5^2 \times 12$ $(= 300\pi$ or 942.47...) **[M1]**

Volume of cone $= \frac{1}{3} \times \pi \times 5^2 \times 6$ $(= 50\pi$ or 157.07...) **[M1]**

1100 or 1099.5... to 1099.6 **[A1]**

(b) Slant height of cone $= \sqrt{5^2 + 6^2}$ $(= \sqrt{61}$ or 7.81...) **[M1]**

Curved surface area of cone $= \pi \times 5 \times$ their $\sqrt{61}$ $(= 122.68...)$ **[M1]**

Curved surface area of cylinder + base $= (2 \times \pi \times 5 \times 12) + (\pi \times 5^2)$ $(= 145\pi$ or 455.53...) **[M1]**

578 or 578.2... **[A1]**

[Total Marks: 7]

6 (a) Total height of large cone $= 20 + 10 = 30$ cm **[B1]**

Volume of large cone $= \frac{1}{3} \times \pi \times 12^2 \times 30$ $(= 1440\pi$ or 4523.89...) **[M1]**

Volume of small cone $= \frac{1}{3} \times \pi \times 4^2 \times 10$ $(= 53.33\pi$ or 167.55...) **[M1]**

4360 or 4356... **[A1]**

(b) Slant height of large cone, $L = \sqrt{12^2 + 30^2}$ $(= \sqrt{1044}$ or 32.31...) AND slant height of small cone, $l = \sqrt{4^2 + 10^2}$ $(= \sqrt{116}$ or 10.77...) **[M1]**

Curved surface area of frustum $= \pi \times 12 \times \sqrt{1044} - \pi \times 4 \times \sqrt{116}$ $(= 1082.7...)$ **[M1]**

Total surface area $= 1082.7... + \pi \times 12^2 + \pi \times 4^2$ **[M1]**

1585 or 1585.3... to 1585.4 **[A1]**

[Total Marks: 8]